Title

Roman to Integer Conversion Using Packages in Java

Objective

To create a package named `number`, define a `Roman` class containing the `romanToInteger()` method, and import and access that method from another Java class.

System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse

Software Requirements

- Java JDK
- VS Code / Eclipse / IntelliJ IDEA
- Windows/Linux operating system

Algorithm / Logic Description

1. Start the program.
2. Create a package named `number`.
3. Define a public class named `Roman` inside the package.
4. Implement the `romanToInteger()` method to convert Roman numerals into integers.
5. Assign values to Roman symbols: I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, and M = 1000.
6. Traverse each character in the Roman numeral.
7. If the current symbol has a smaller value than the next symbol, subtract its value.
8. Otherwise, add its value to the result.
9. Create another class named `Main`.
10. Import the `Roman` class using `import number.Roman;`.
11. Read a Roman numeral from the user.
12. Call `Roman.romanToInteger()` and display the integer value.
13. Stop the program.

Package Structure

```
Project
│
├── number
│   └── Roman.java
│
└── Main.java
```

Source Code

```
 package number;


public class Roman {

    public static int romanToInteger(String s) {

        int result = 0;


        for (int i = 0; i < s.length(); i++) {

            int value;


            switch (s.charAt(i)) {

                case 'I': value = 1; break;

                case 'V': value = 5; break;

                case 'X': value = 10; break;

                case 'L': value = 50; break;

                case 'C': value = 100; break;

                case 'D': value = 500; break;

                case 'M': value = 1000; break;

                default: value = 0;
```

```
    }

    if (i + 1 < s.length()) {
        int next = 0;

        switch (s.charAt(i + 1)) {
            case 'I': next = 1; break;
            case 'V': next = 5; break;
            case 'X': next = 10; break;
            case 'L': next = 50; break;
            case 'C': next = 100; break;
            case 'D': next = 500; break;
            case 'M': next = 1000; break;
        }

        if (value < next) {
            result -= value;
        } else {
            result += value;
        }
    } else {
        result += value;
    }
}
```

```
        return result;
    }
}

import java.util.Scanner;
import number.Roman;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roman numeral: ");
        String s = sc.nextLine();

        int result = Roman.romanToInteger(s);

        System.out.println("Integer value: " + result);

        sc.close();
    }
}
```

Compilation and Execution Commands

Run these commands from the project directory containing `Main.java` and the `number` folder.

Compile:

javac number/Roman.java Main.java

Execute:

java Main

Output Screen

Enter Roman numeral: LVIII
Integer value: 58

Result

The Java program was successfully implemented by defining the `number` package, creating the `Roman` class, and importing the class explicitly into another Java program to convert Roman numerals into integers.